BLDEA'S
COMMERCE BHS ARTS AND TGP SCIENCE COLLEGE, JAMAKHANDI

BCA VI SEMESTER 2025-26

PROJECT SYNOPSIS

Title of the Project

SMART COLLEGE COMPANION

Degree: Bachelor of Computer Applications (BCA)

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Submitted By

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SMART COLLEGE COMPANION

1. Introduction and Objectives of the Project

In today's digital era, educational institutions require efficient and reliable systems to manage academic, administrative, and student-related activities. Traditional college management methods such as manual attendance registers, notice boards, paper-based assignments, and offline communication often result in delays, data redundancy, and human errors.

The Smart College Companion is a web-based application developed using the Django framework that provides a centralized digital platform for students, faculty, and administrators. The system automates academic processes such as student management, attendance tracking, assignment submission and notifications.

Objectives:

- To digitize and automate college academic and administrative processes
- To reduce paperwork and manual data handling
- To provide centralized access to student and academic information
- To improve communication between students, faculty, and administration
- To ensure data accuracy and minimize human errors
- To provide real-time updates and notifications

2. Existing System

The existing system followed by many colleges is either fully manual or partially computerized. Attendance is recorded in registers, assignments are submitted physically, and notices are displayed on notice boards. This system is time-consuming, error-prone, and does not provide a centralized data repository.

- Students depend on notice boards, timetable, verbal announcements, or multiple apps/websites for information.
- Attendance records are often maintained manually or in separate systems, making it difficult for students to track their attendance.

- Assignment submission and evaluation processes are not fully automated.
- Important notices and announcements may be missed due to lack of real- time notification.

3.Proposed System

The Smart College Companion is a fully digital, centralized, and intelligent college management system developed to overcome the limitations of the existing system.

The proposed Smart College Companion system replaces traditional processes with a centralized web application. All academic and administrative activities are managed digitally through structured database tables. The system ensures secure login, online assignment submission, attendance tracking, referring study materials, conduct exams and upload each student's scores and instant notifications.

The system provides a single centralized platform for students, teachers, and administrators.

Students can access:

- Notices and announcements
- Timetables and exam schedules
- Attendance records
- Assignment uploads and deadlines
- Study Materials and exam scores

Teachers can:

- Upload attendance, study materials and assignments
- Post notices and exam information with exam scores

Admin can:

- Manage users (students and teachers)
- Control and update college, department and staff informations

- Monitor overall system activity
- Real-time notifications ensure students never miss important updates.
- Data is stored securely and can be accessed anytime.
- Create Timetable which is both staff and students friendly

The proposed system enhances the overall academic experience by providing fast, accurate, and reliable information to all users.

4. Project Category

Web Application Development

5. Models in the Project

- **User Authentication Model:** Manages secure login and role-based access for students, faculty, and administrators using the userlogin table.
- **Student Profile Model:** Maintains student personal and academic details such as department, semester, year, email, and mobile number using the studentdetails table.
- **Subject Management Model:** Handles Subject details and student enrollment using the Subject table.
- **Timetable Management Model:** Manages daily class schedules including subjects, instructors, days, and timings using the timetable table.
- **Attendance Management Model:** Records and tracks student attendance for each subject using the attendance table.
- **Assignment Management Model:** Manages assignment creation, submission, and evaluation using assignments, submissions, and feedback tables.

- **Notification and Announcement Model:** Displays important academic updates using notifications and announcements tables.

- **Exam and Exam scores Model:** Designed to manage examination details and track student performance efficiently.

- **Instructor & Department Model:** Manages faculty information and organizes them under specific academic departments.

- **Study Materials Model:** Designed to manage and distribute academic resources to students in a centralized and organized manner such as notes, PDFs, presentations, and other learning resources.

- **Feedback Model:** Enables collect and manage feedback from students regarding instructors, courses, or the overall system.

6.Tools and Technologies

Component	Technology
Frontend	HTML5, CSS3, JavaScript
Backend	Python 3.13 / 3.14, Django Framework 4.x
Database	SQLite (Development), MySQL
IDE	Visual Studio Code, PyCharm

7.Hardware and Software Requirements

Hardware Requirements:

Processor : Intel i3 or above

RAM : Minimum 4GB

Storage : Minimum 20GB free space

Software Requirements:

Operating System: Windows / Linux

Web Browser : Google Chrome / Mozilla Firefox

Python3.13 /3.14, Django Framework